Reg. No. | | | | | | | | | | | [18CSOE01]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

DATA STRUCTURES AND ALGORITHMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Discuss on big oh (O), Big omega (Ω) and big theta (Θ) notations.	**06**
	b.	Analyze the time complexity of Iterative and Recursive Algorithms with a suitable example.	**08**
2.	**a.**	Design the pseudo code for array implementation of Circular Queue.	**07**
	b.	Implement a C program to push and pop using stack.	**07**
3.	**a.**	Define Binary Tree? Explain the types of Binary tree and discuss the different tree traversal methods.	**07**
	b.	Explain creation and insertion Operations of Binary Search Tree with Example.	**07**
		OR	
4.	**a.**	Implement tree traversal methods with C function.	**07**
	b.	What are priority queues? List the application of priority queues and show how it can be implemented.	**07**
5.	**a.**	Write an algorithm for heap sort and Convert the following array to a max heap 16,4,7,112,19	**07**
	b.	Explain the working of DFS algorithm with a suitable example.	**07**
		OR	
6.	**a.**	Write Merge sort algorithm and explain with suitable example.	**07**
	b	Discuss the working of Quick sort algorithm.	**07**
7.	**a.**	Define Graph? Explain the different ways of representing graph. Also give the application of Graph.	**07**
	b.	Discuss the different Graph Traversal Techniques with example.	**07**
		OR	
8.	**a.**	Write the Kruskal's algorithm for finding the minimum spanning tree.	**07**
	b.	Formulate an algorithm to find the shortest path using Dijkstra's algorithm and explain with example.	**07**